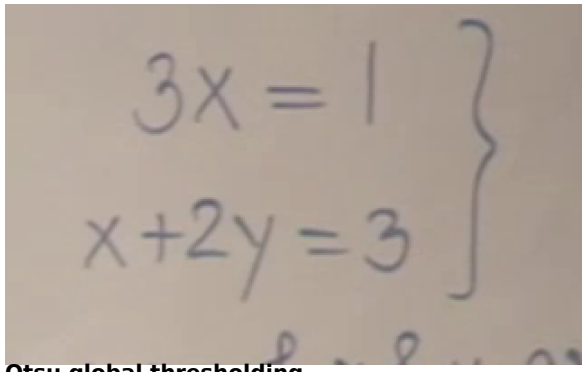
Actual lecture: technique comparison

lecture_01_10423 | Native 1920 x 1080 | Source crop (290, 430, 320, 200)

Same crop and scale throughout; nearest-neighbor 2x raster; no display contrast correction.

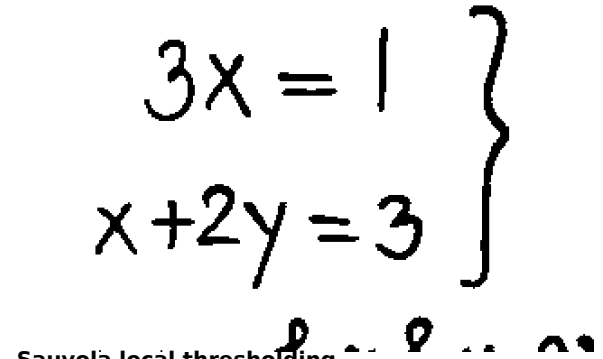
Original RGB

Unprocessed lecture frame



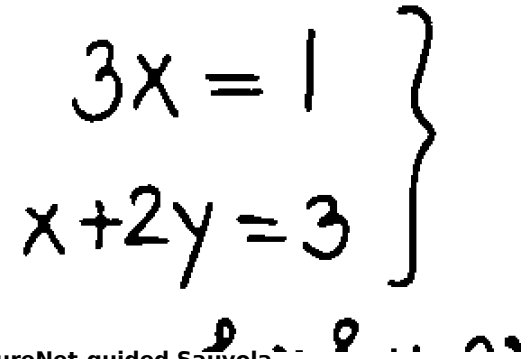
Reference annotation

Public pixel ground truth



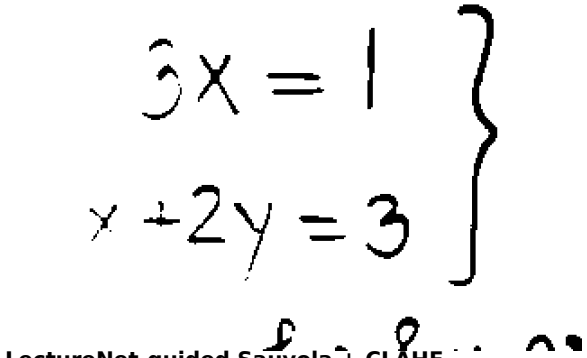
FCN-LectureNet V34

Original 8-bit threshold = 128



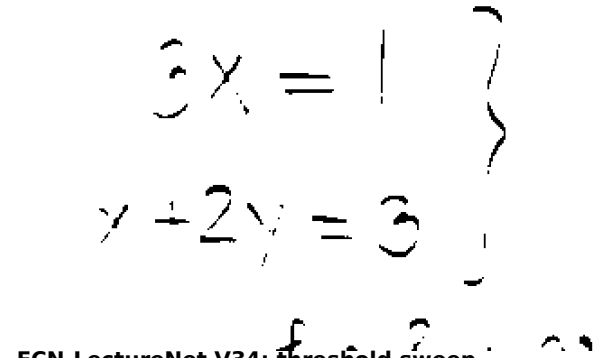
Otsu global thresholding

Grayscale; Gaussian blur disabled



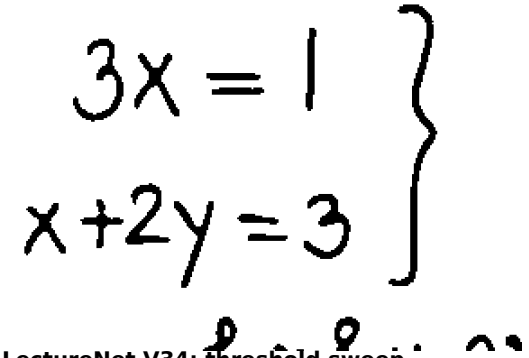
Sauvola local thresholding

Window = 25; k = 0.2; CLAHE disabled



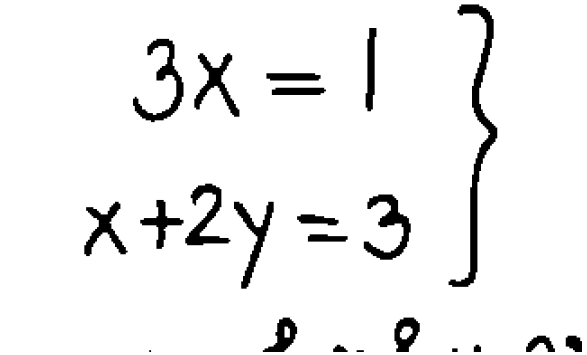
LectureNet-guided Sauvola

Window = 25; k = 0.2; dilation radius = 2



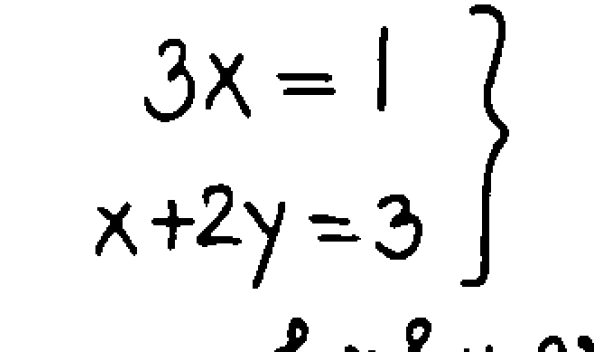
LectureNet-guided Sauvola + CLAHE

Window = 25; k = 0.2; dilation radius = 2



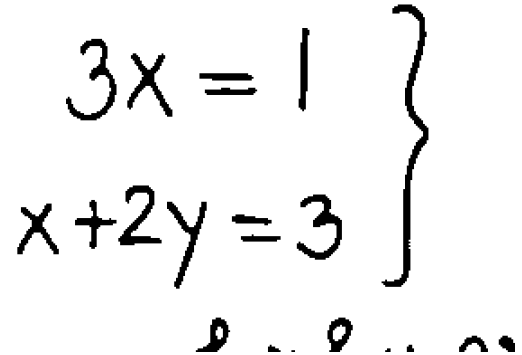
FCN-LectureNet V34: threshold sweep

Writing probability threshold = 0.35



FCN-LectureNet V34: threshold sweep

Writing probability threshold = 0.65

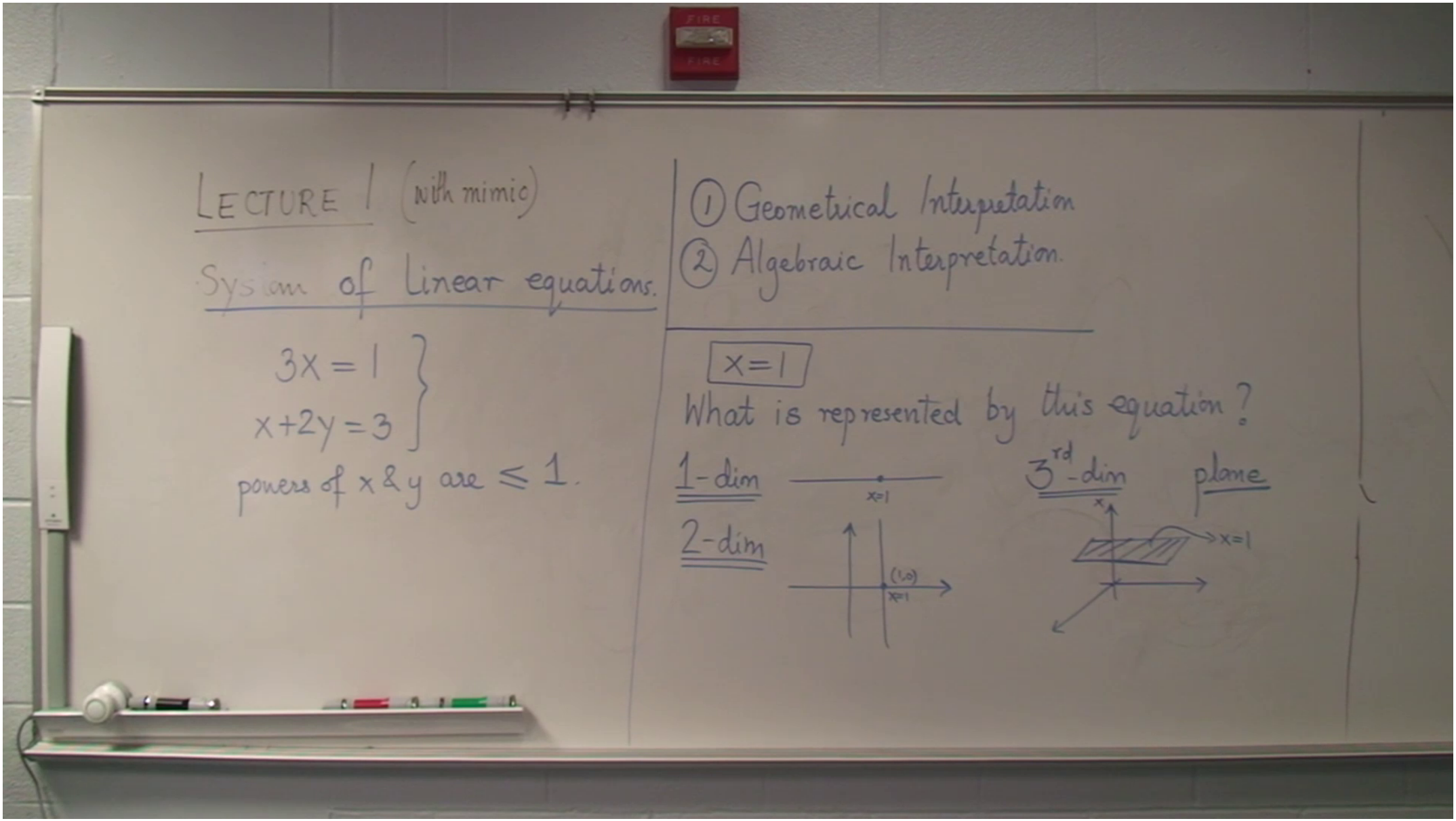


Only FCN-LectureNet is a trained neural network; guided Sauvola is a refinement, not a separately trained model.

Original RGB

Unprocessed lecture frame

Real AccessMath2015 | lecture_01_10423 | Native 1920 x 1080 | No smoothing



Same full-frame placement on every page. Three frames only; mathematical OCR accuracy was not evaluated.

Reference annotation

Public pixel ground truth

Real AccessMath2015 | lecture_01_10423 | Native 1920 x 1080 | No smoothing

LECTURE 1 (with mimio)

System of Linear equations.

$$\left.\begin{aligned} 3x = 1 \\ x+2y = 3 \end{aligned}\right\}$$

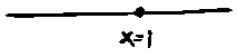
powers of x & y are ≤ 1.

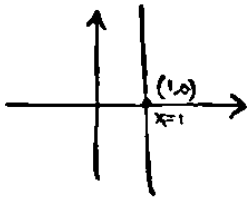
① Geometrical Interpretation

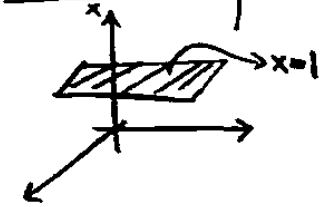
② Algebraic Interpretation.

$x=1$

What is represented by this equation?

1-dim 

2-dim 

3rd-dim plane 

Same full-frame placement on every page. Three frames only; mathematical OCR accuracy was not evaluated.

FCN-LectureNet V34

Original 8-bit threshold = 128

Real AccessMath2015 | lecture_01_10423 | Native 1920 x 1080 | No smoothing

LECTURE 1 (with mimic)

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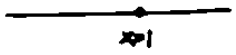
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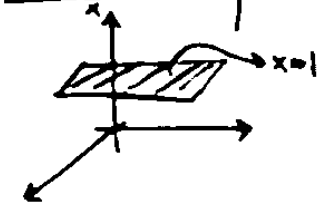
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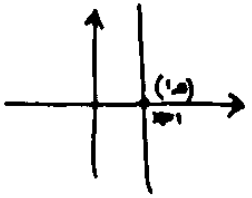
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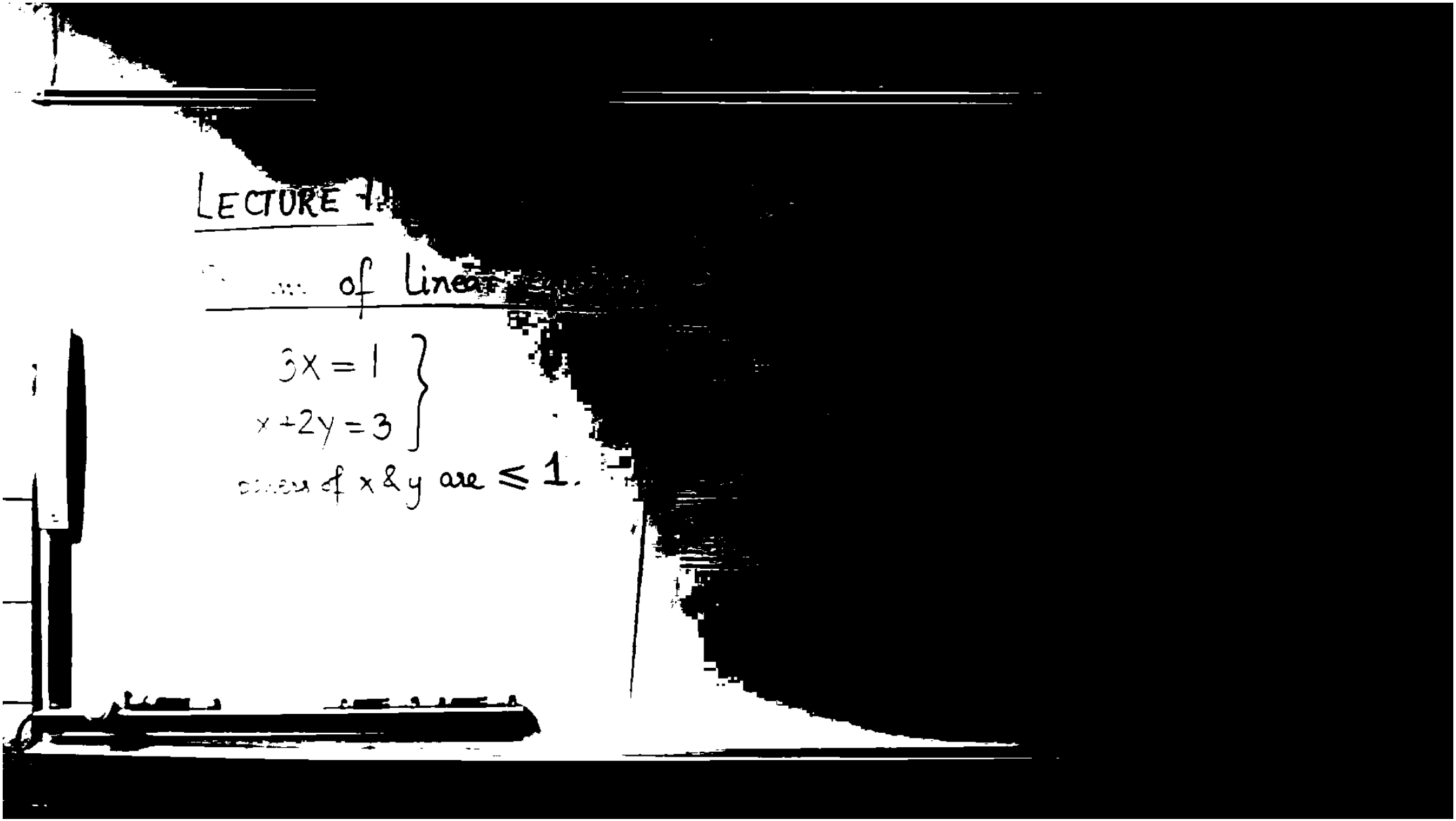
2-dim 

Same full-frame placement on every page. Three frames only; mathematical OCR accuracy was not evaluated.

Otsu global thresholding

Grayscale; Gaussian blur disabled

Real AccessMath2015 | lecture_01_10423 | Native 1920 x 1080 | No smoothing

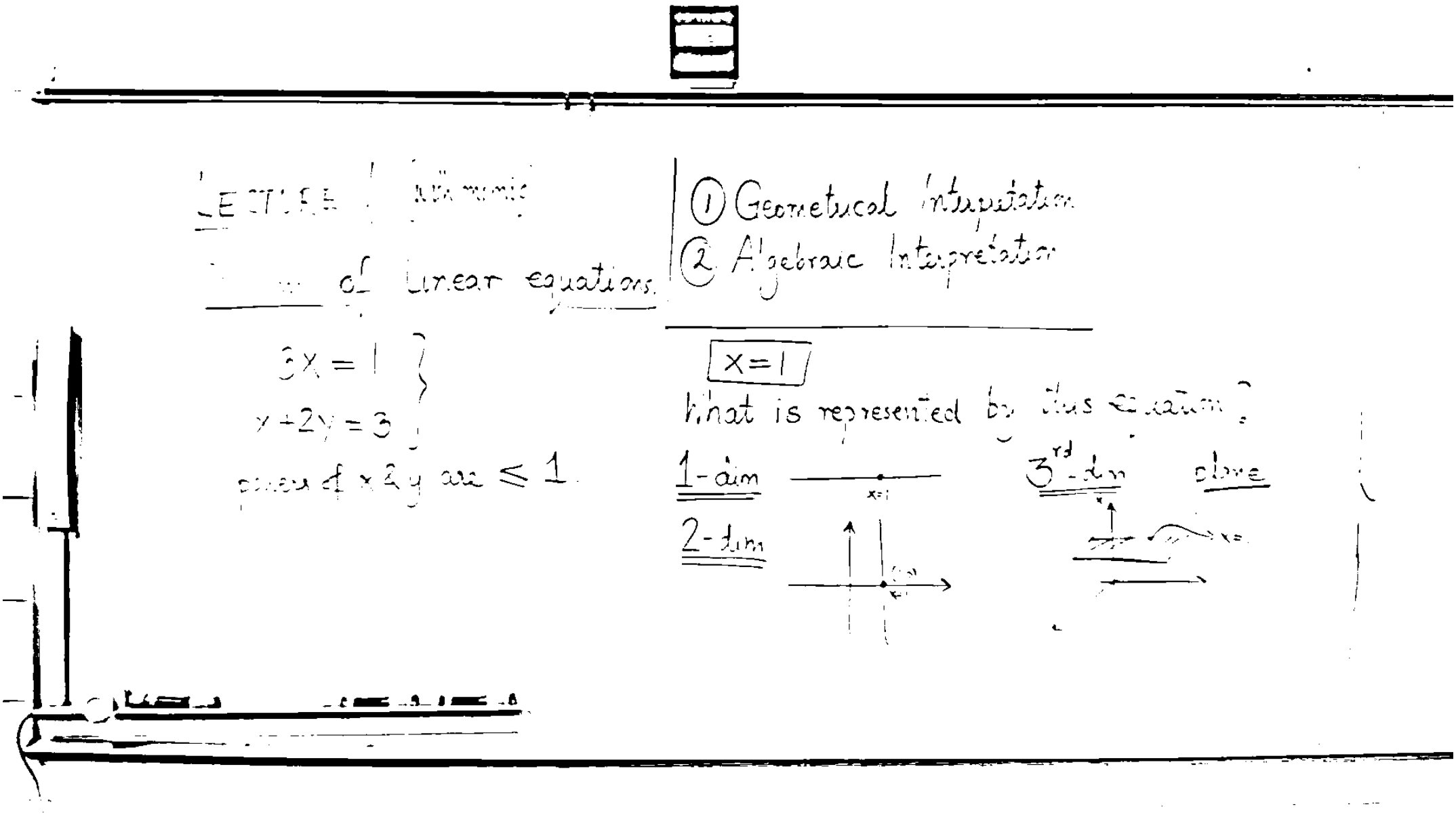


Same full-frame placement on every page. Three frames only; mathematical OCR accuracy was not evaluated.

Sauvola local thresholding

Window = 25; k = 0.2; CLAHE disabled

Real AccessMath2015 | lecture_01_10423 | Native 1920 x 1080 | No smoothing



Same full-frame placement on every page. Three frames only; mathematical OCR accuracy was not evaluated.

LectureNet-guided Sauvola

Window = 25; k = 0.2; dilation radius = 2

Real AccessMath2015 | lecture_01_10423 | Native 1920 x 1080 | No smoothing

LECTURE 1 (with mimic)

System of linear equations.

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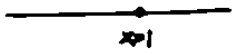
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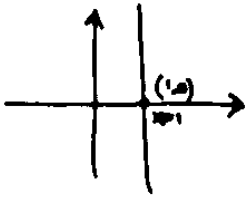
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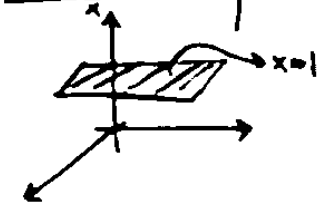
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LectureNet-guided Sauvola + CLAHE

Window = 25; k = 0.2; dilation radius = 2

Real AccessMath2015 | lecture_01_10423 | Native 1920 x 1080 | No smoothing

1

LECTURE 1 (with mimic)

System of linear equations.

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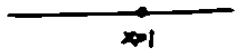
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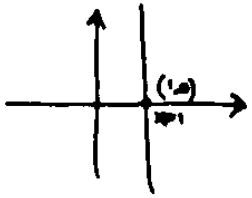
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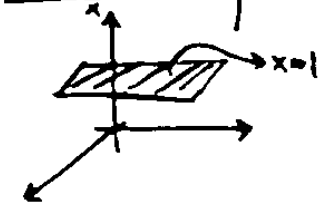
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FCN-LectureNet V34: threshold sweep

Writing probability threshold = 0.35

Real AccessMath2015 | lecture_01_10423 | Native 1920 x 1080 | No smoothing

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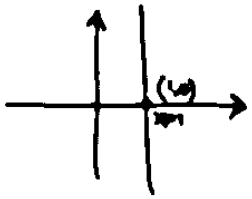
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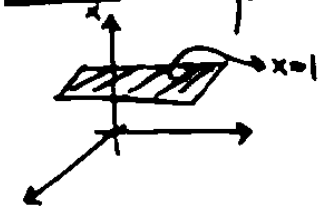
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Real AccessMath2015 | lecture_01_10423 | Native 1920 x 1080 | No smoothing

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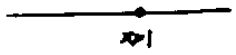
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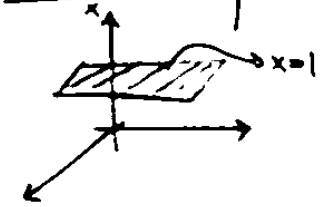
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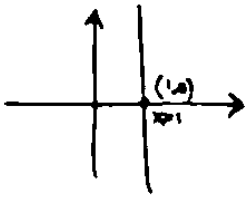
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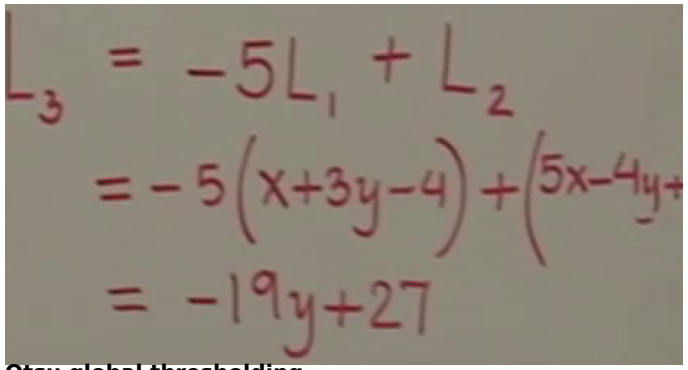
Actual lecture: technique comparison

lecture_02_44601 | Native 1920 x 1080 | Source crop (1320, 480, 470, 250)

Same crop and scale throughout; nearest-neighbor 2x raster; no display contrast correction.

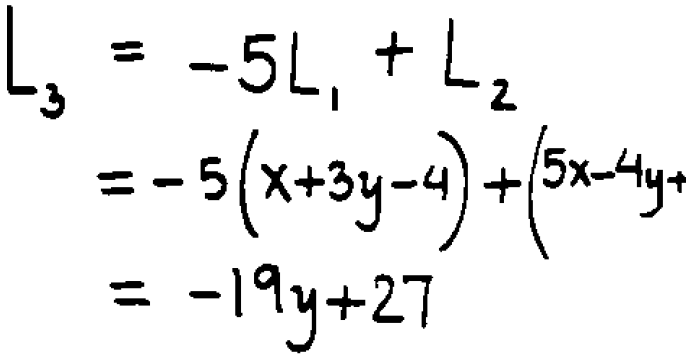
Original RGB

Unprocessed lecture frame



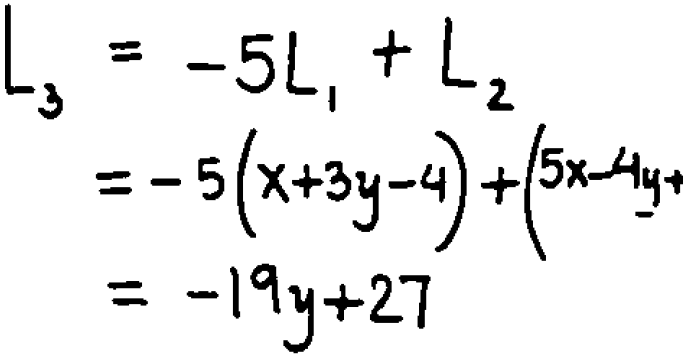
Reference annotation

Public pixel ground truth



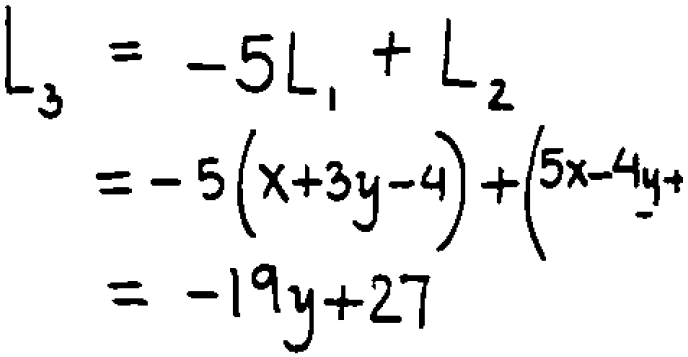
FCN-LectureNet V34

Original 8-bit threshold = 128



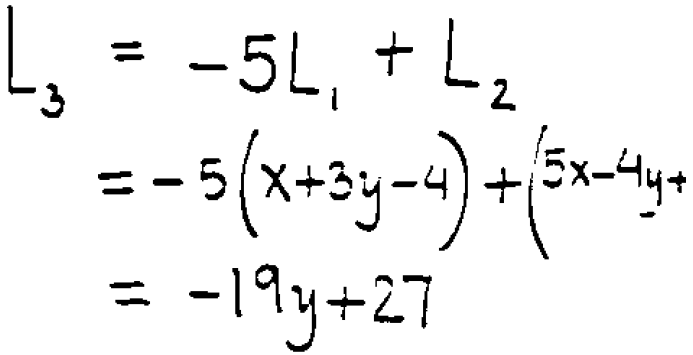
Otsu global thresholding

Grayscale; Gaussian blur disabled



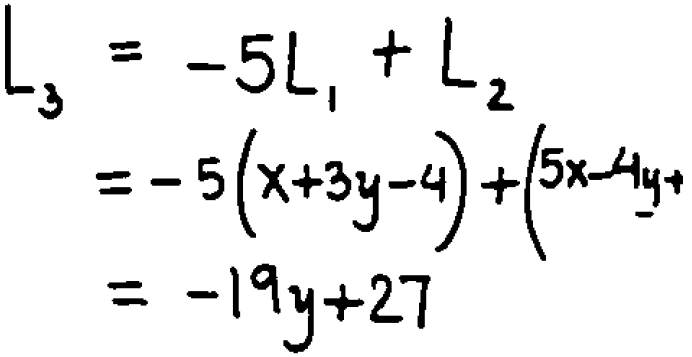
Sauvola local thresholding

Window = 25; k = 0.2; CLAHE disabled



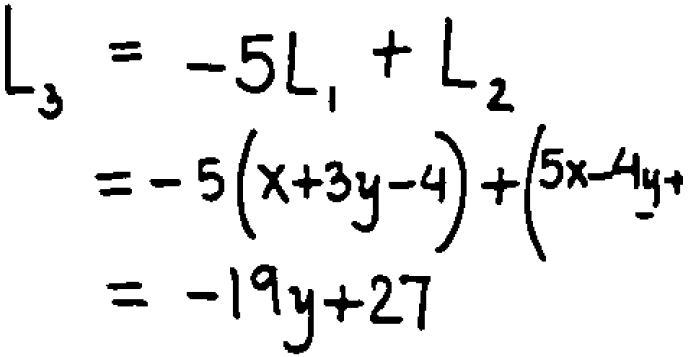
LectureNet-guided Sauvola

Window = 25; k = 0.2; dilation radius = 2



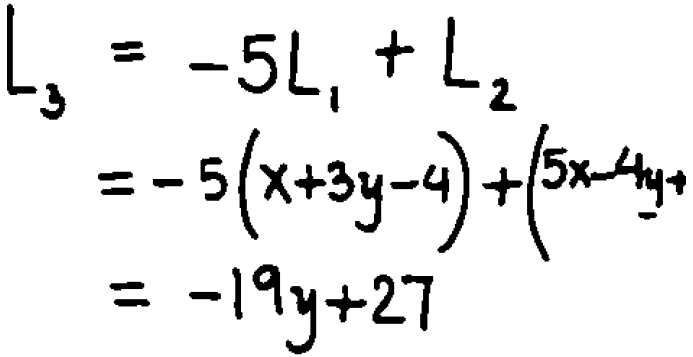
LectureNet-guided Sauvola + CLAHE

Window = 25; k = 0.2; dilation radius = 2



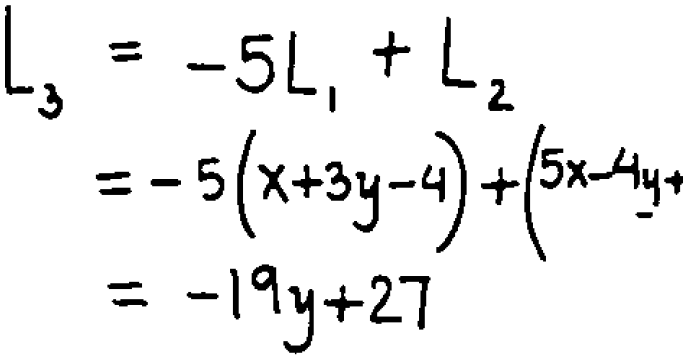
FCN-LectureNet V34: threshold sweep

Writing probability threshold = 0.35



FCN-LectureNet V34: threshold sweep

Writing probability threshold = 0.65

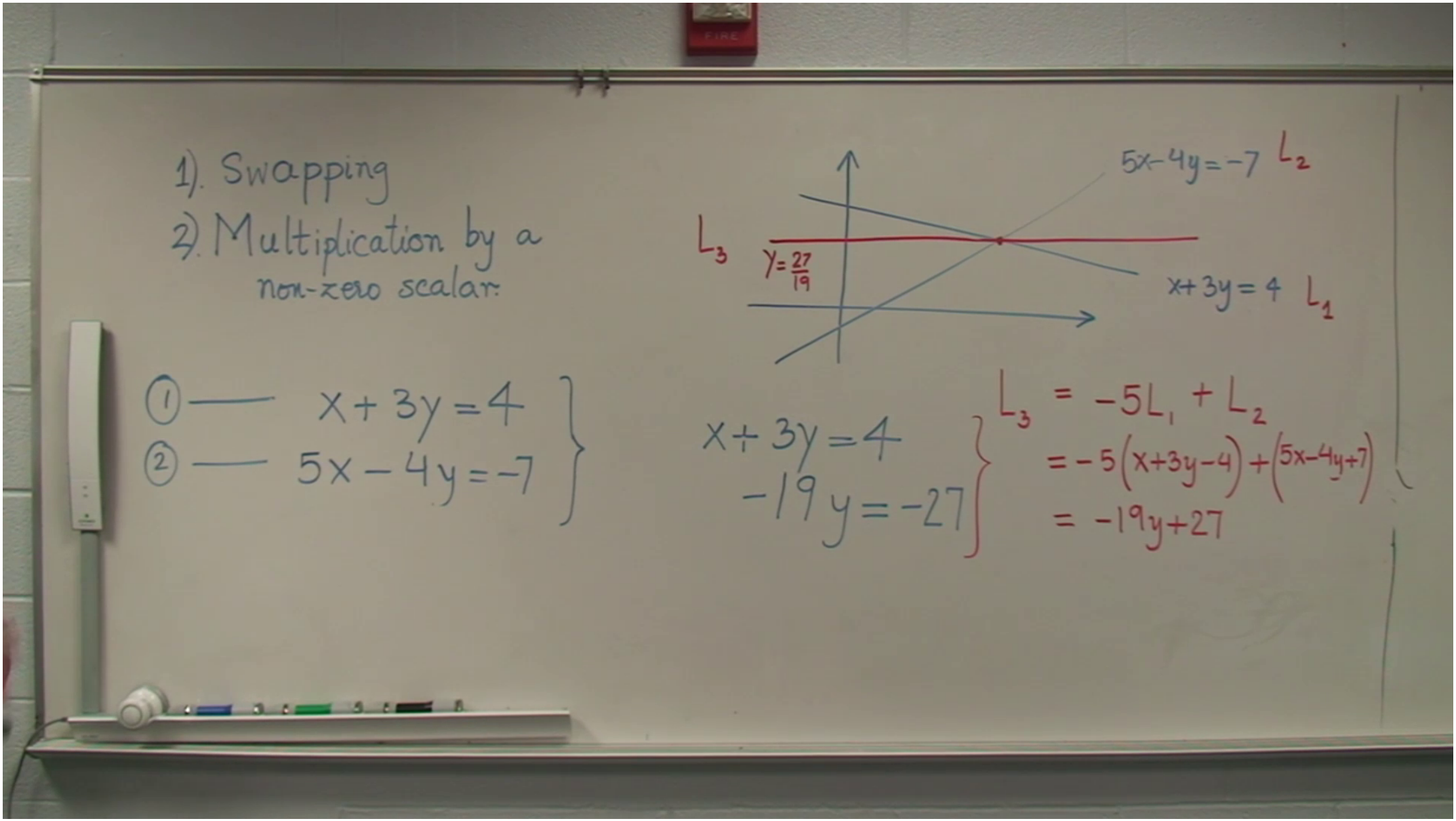


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Original RGB

Unprocessed lecture frame

Real AccessMath2015 | lecture_02_44601 | Native 1920 x 1080 | No smoothing



Same full-frame placement on every page. Three frames only; mathematical OCR accuracy was not evaluated.

Reference annotation

Public pixel ground truth

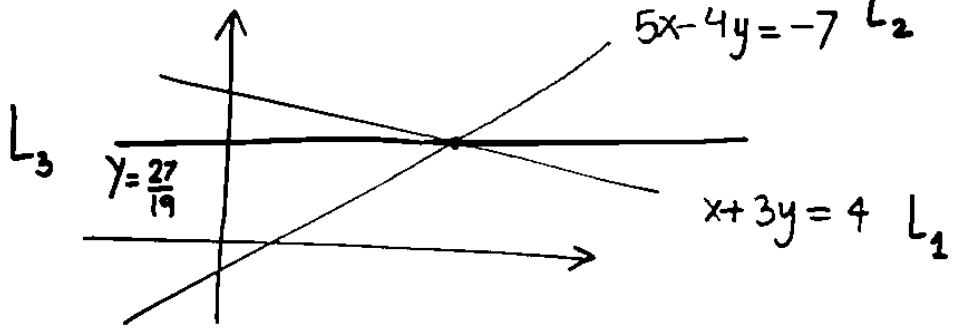
Real AccessMath2015 | lecture_02_44601 | Native 1920 x 1080 | No smoothing

1). Swapping

2). Multiplication by a non-zero scalar.

① —— $x+3y=4$

② —— $5x-4y=-7$



$$\begin{matrix} x+3y=4 \\ -19y=-27 \end{matrix}$$

$$\begin{aligned} L_3 &= -5L_1 + L_2 \\ &= -5(x+3y-4)+(5x-4y+7) \\ &= -19y+27 \end{aligned}$$

Same full-frame placement on every page. Three frames only; mathematical OCR accuracy was not evaluated.

FCN-LectureNet V34

Original 8-bit threshold = 128

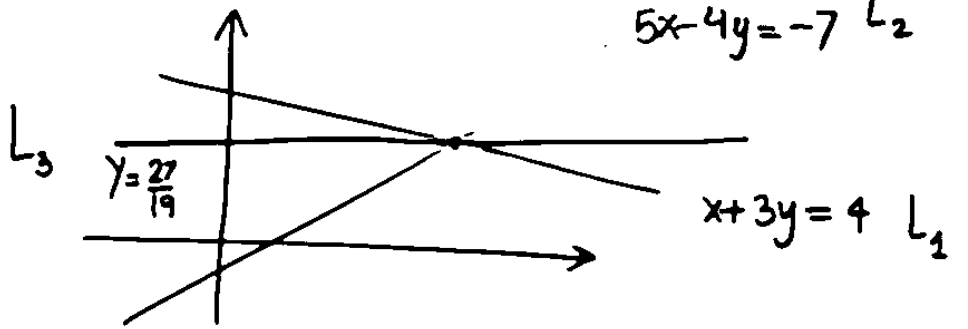
Real AccessMath2015 | lecture_02_44601 | Native 1920 x 1080 | No smoothing

1). Swapping

2). Multiplication by a non-zero scalar:

① —— $x+3y=4$

② —— $5x-4y=-7$



$x+3y=4$

$-19y=-27$

$L_3 = -5L_1 + L_2$

$= -5(x+3y-4) + (5x-4y+7)$

$= -19y+27$

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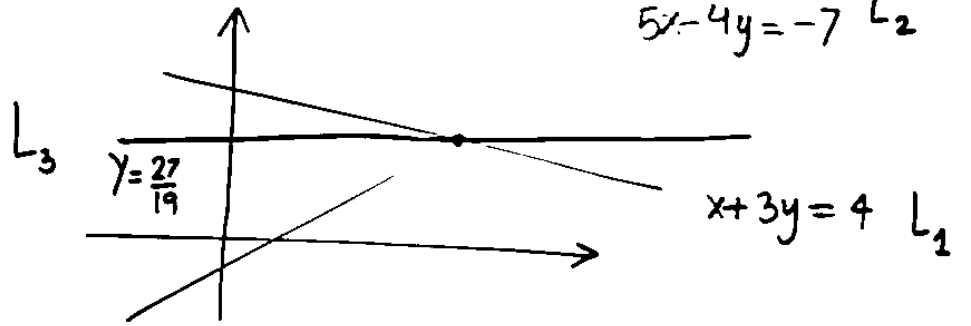
Otsu global thresholding

Grayscale; Gaussian blur disabled

Real AccessMath2015 | lecture_02_44601 | Native 1920 x 1080 | No smoothing

1) Swapping

2) Multiplication by a non-zero scalar



① —— $x+3y=4$

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$x+3y=4$

$-19y=-27$

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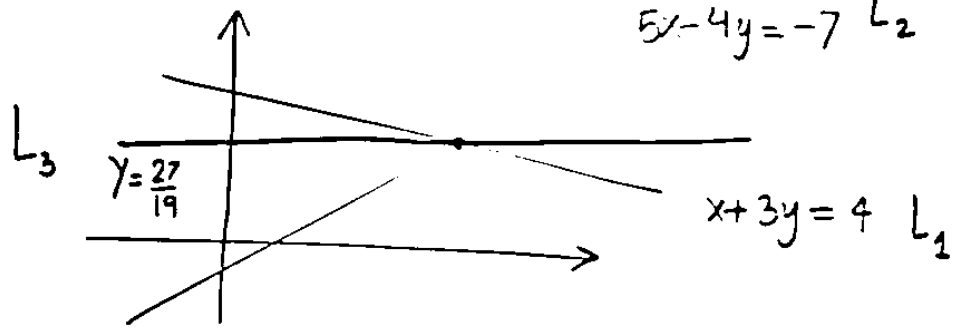
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Real AccessMath2015 | lecture_02_44601 | Native 1920 x 1080 | No smoothing

1) Swapping
2) Multiplication by a non-zero scalar

$$\begin{cases} \text{①} \quad x+3y=4 \\ \text{②} \quad 5x-4y=-7 \end{cases}$$



$$\begin{cases} x+3y=4 \\ -19y=-27 \end{cases}$$

$$\begin{aligned} L_3 &= -5L_1 + L_2 \\ &= -5(x+3y-4)+(5x-4y+7) \\ &= -19y+27 \end{aligned}$$

Same full-frame placement on every page. Three frames only; mathematical OCR accuracy was not evaluated.

LectureNet-guided Sauvola

Window = 25; k = 0.2; dilation radius = 2

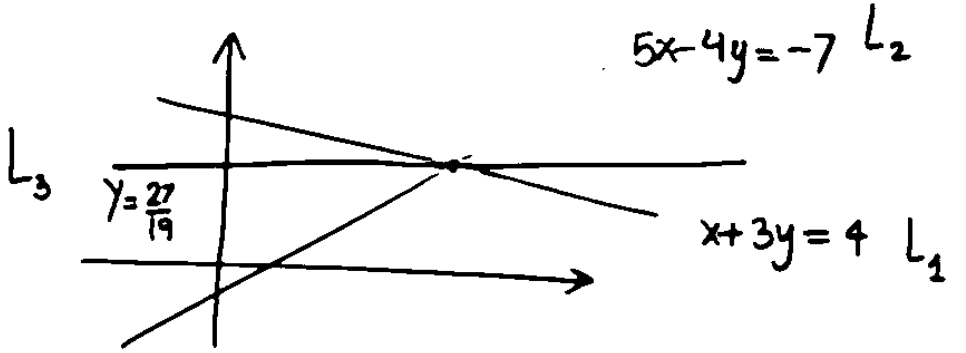
Real AccessMath2015 | lecture_02_44601 | Native 1920 x 1080 | No smoothing

1). Swapping

2). Multiplication by a non-zero scalar:

① —— $x+3y=4$

② —— $5x-4y=-7$



$x+3y=4$

$-19y=-27$

$L_3 = -5L_1 + L_2$

$= -5(x+3y-4) + (5x-4y+7)$

$= -19y+27$

Same full-frame placement on every page. Three frames only; mathematical OCR accuracy was not evaluated.

LectureNet-guided Sauvola + CLAHE

Window = 25; k = 0.2; dilation radius = 2

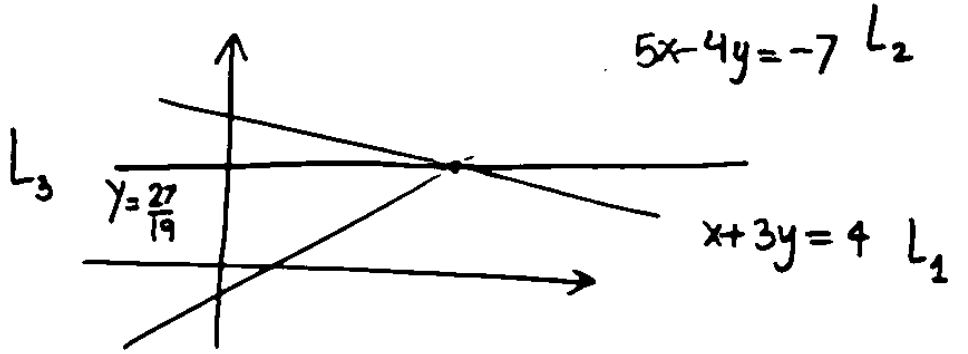
Real AccessMath2015 | lecture_02_44601 | Native 1920 x 1080 | No smoothing

1). Swapping

2). Multiplication by a non-zero scalar:

① —— $x+3y=4$

② —— $5x-4y=-7$



$$\begin{aligned} x+3y&=4 \\ -19y&=-27 \end{aligned}$$

$$\begin{aligned} L_3 &= -5L_1 + L_2 \\ &= -5(x+3y-4)+(5x-4y+7) \\ &= -19y+27 \end{aligned}$$

Same full-frame placement on every page. Three frames only; mathematical OCR accuracy was not evaluated.

FCN-LectureNet V34: threshold sweep

Writing probability threshold = 0.35

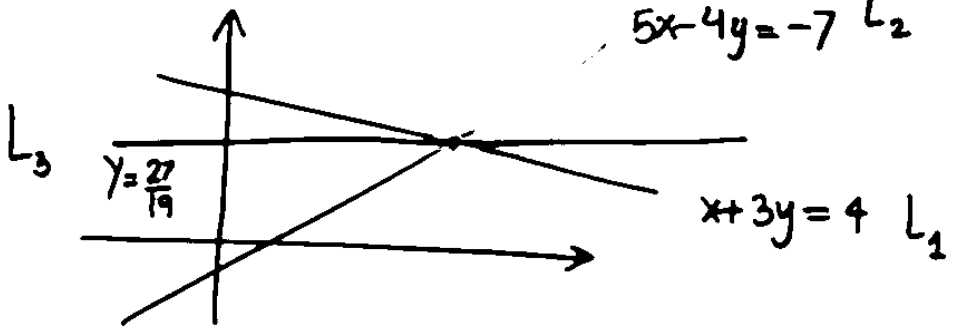
Real AccessMath2015 | lecture_02_44601 | Native 1920 x 1080 | No smoothing

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FCN-LectureNet V34: threshold sweep

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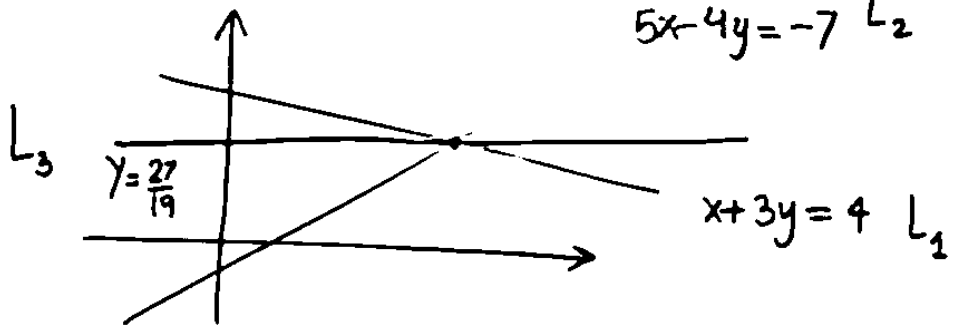
Real AccessMath2015 | lecture_02_44601 | Native 1920 x 1080 | No smoothing

1). Swapping

2). Multiplication by a non-zero scalar:

① —— $x+3y=4$

② —— $5x-4y=-7$



$x+3y=4$

$-19y=-27$

$L_3 = -5L_1 + L_2$

$= -5(x+3y-4) + (5x-4y+7)$

$= -19y+27$

Same full-frame placement on every page. Three frames only; mathematical OCR accuracy was not evaluated.

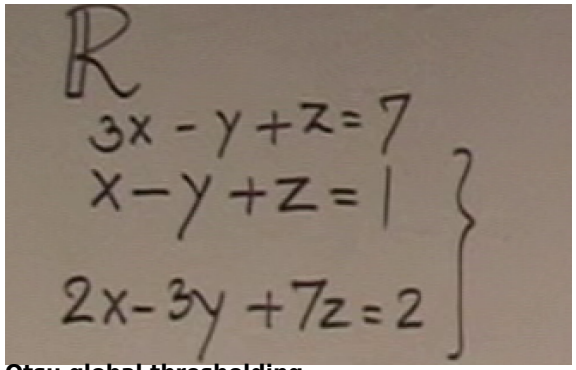
Actual lecture: technique comparison

nm_lecture_01_19899 | Native 1920 x 1080 | Source crop (150, 435, 330, 210)

Same crop and scale throughout; nearest-neighbor 2x raster; no display contrast correction.

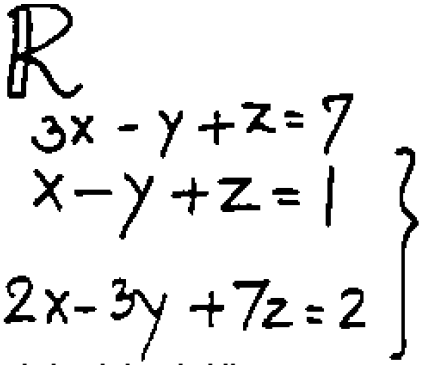
Original RGB

Unprocessed lecture frame



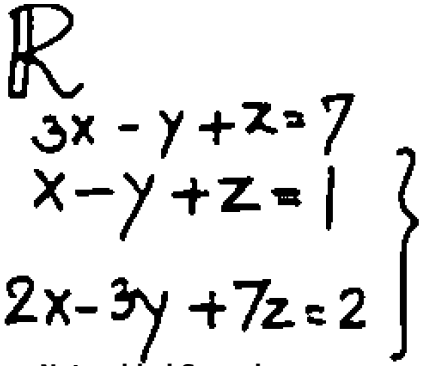
Reference annotation

Public pixel ground truth



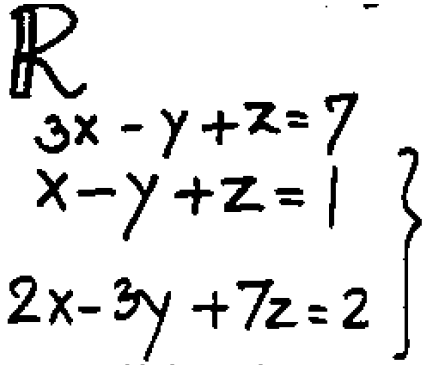
FCN-LectureNet V34

Original 8-bit threshold = 128



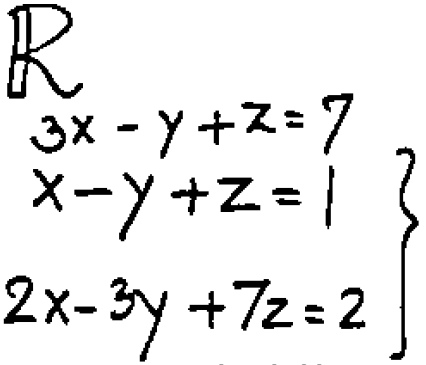
Otsu global thresholding

Grayscale; Gaussian blur disabled



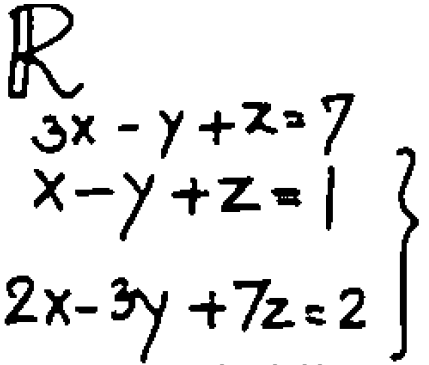
Sauvola local thresholding

Window = 25; k = 0.2; CLAHE disabled



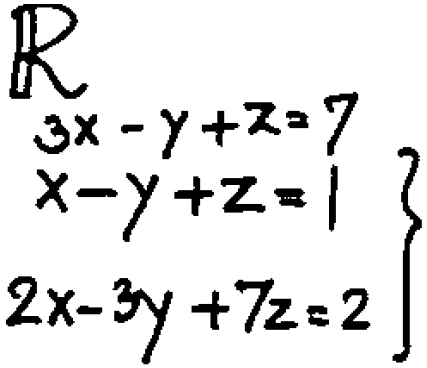
LectureNet-guided Sauvola

Window = 25; k = 0.2; dilation radius = 2



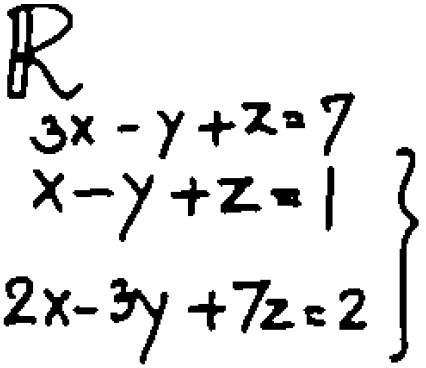
LectureNet-guided Sauvola + CLAHE

Window = 25; k = 0.2; dilation radius = 2



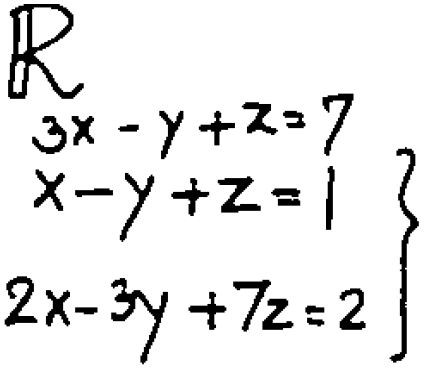
FCN-LectureNet V34: threshold sweep

Writing probability threshold = 0.35



FCN-LectureNet V34: threshold sweep

Writing probability threshold = 0.65

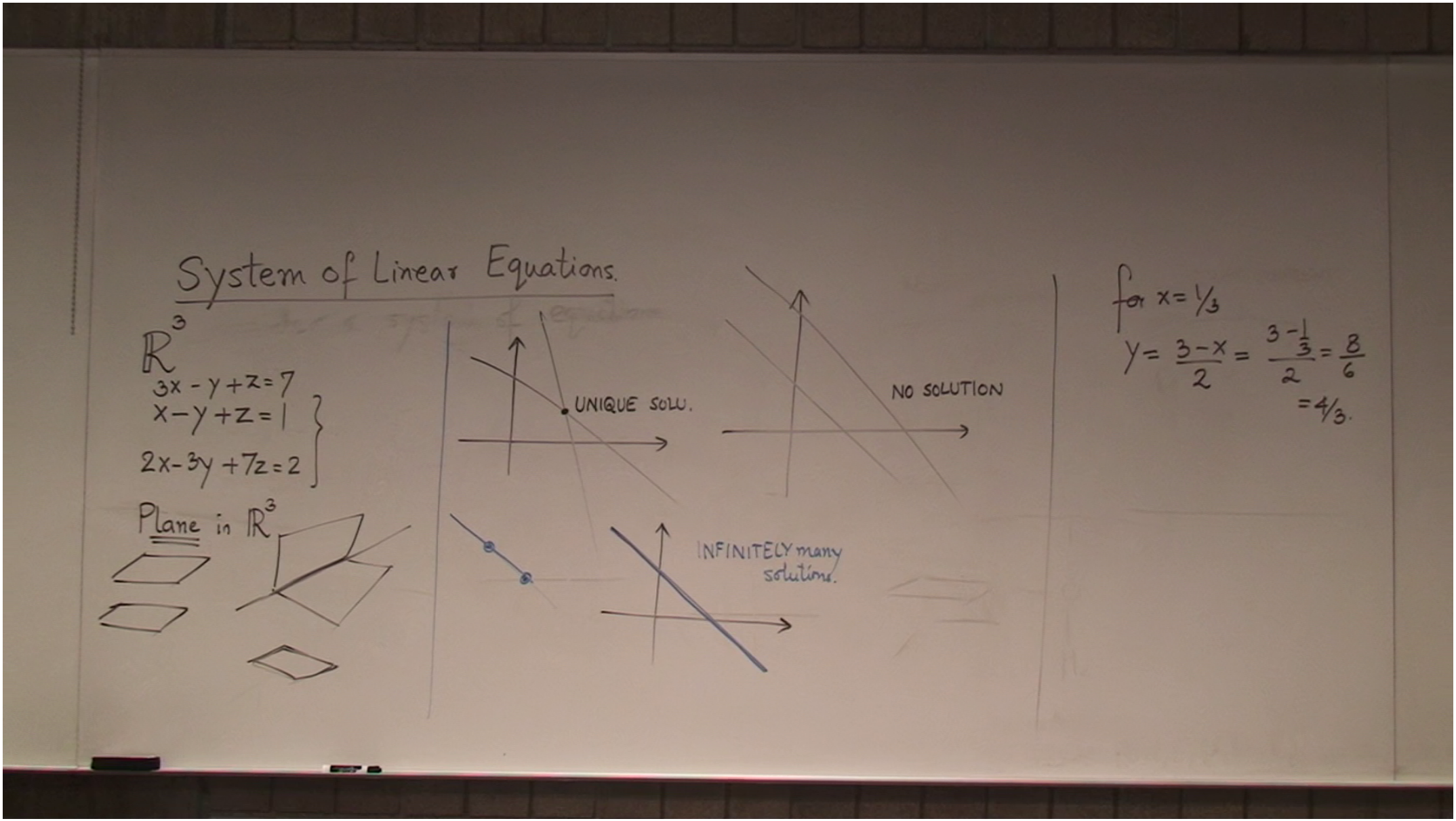


Only FCN-LectureNet is a trained neural network; guided Sauvola is a refinement, not a separately trained model.

Original RGB

Unprocessed lecture frame

Real AccessMath2015 | nm_lecture_01_19899 | Native 1920 x 1080 | No smoothing

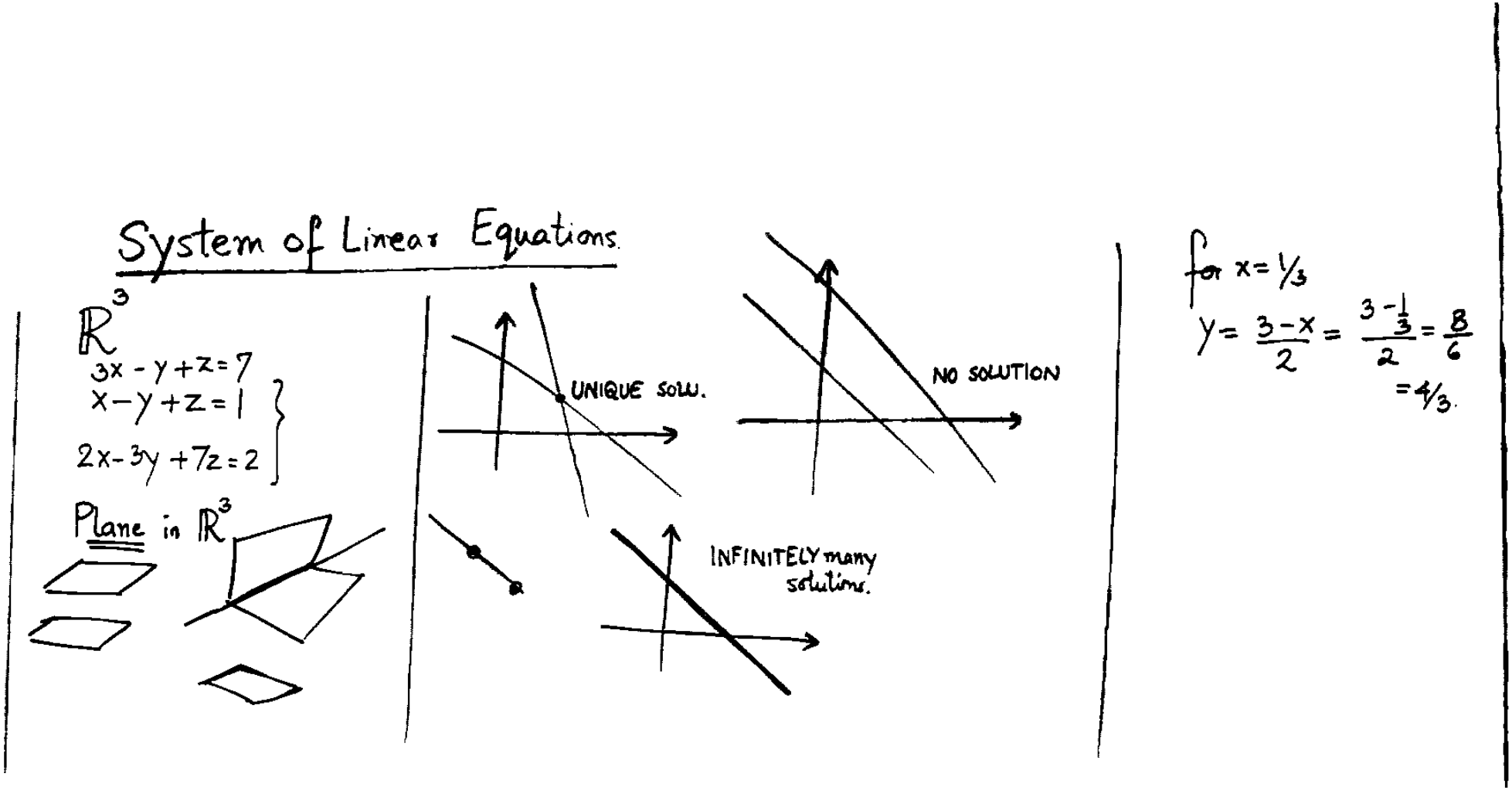


Same full-frame placement on every page. Three frames only; mathematical OCR accuracy was not evaluated.

Reference annotation

Public pixel ground truth

Real AccessMath2015 | nm_lecture_01_19899 | Native 1920 x 1080 | No smoothing



Same full-frame placement on every page. Three frames only; mathematical OCR accuracy was not evaluated.

FCN-LectureNet V34

Original 8-bit threshold = 128

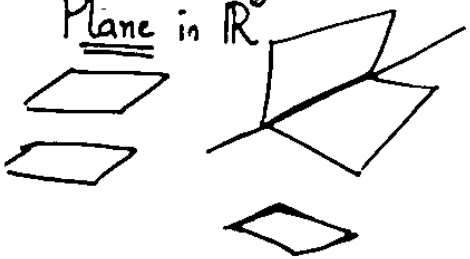
Real AccessMath2015 | nm_lecture_01_19899 | Native 1920 x 1080 | No smoothing

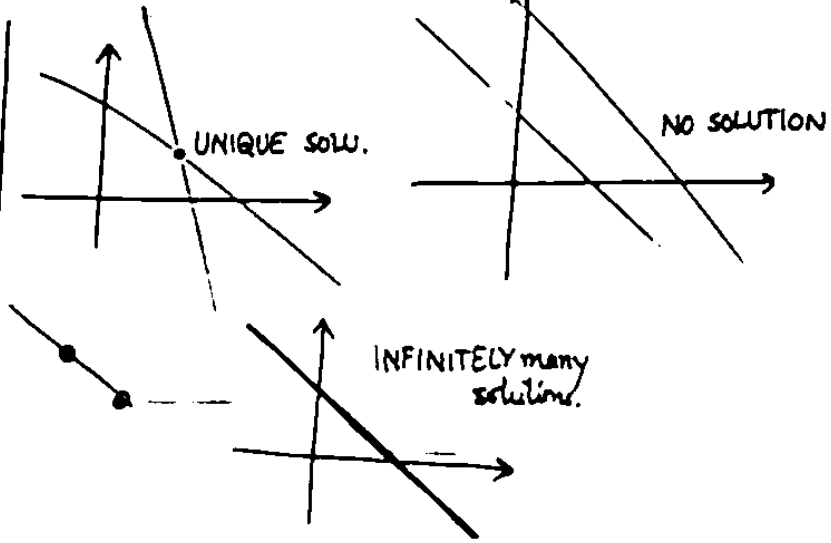
System of Linear Equations.

$\mathbb{R}^3$

$3x - y + z = 7$
$x - y + z = 1$
$2x - 3y + 7z = 2$

Plane in $\mathbb{R}^3$





for $x = 1/3$

$y = \frac{3-x}{2} = \frac{3-\frac{1}{3}}{2} = \frac{8}{6}$

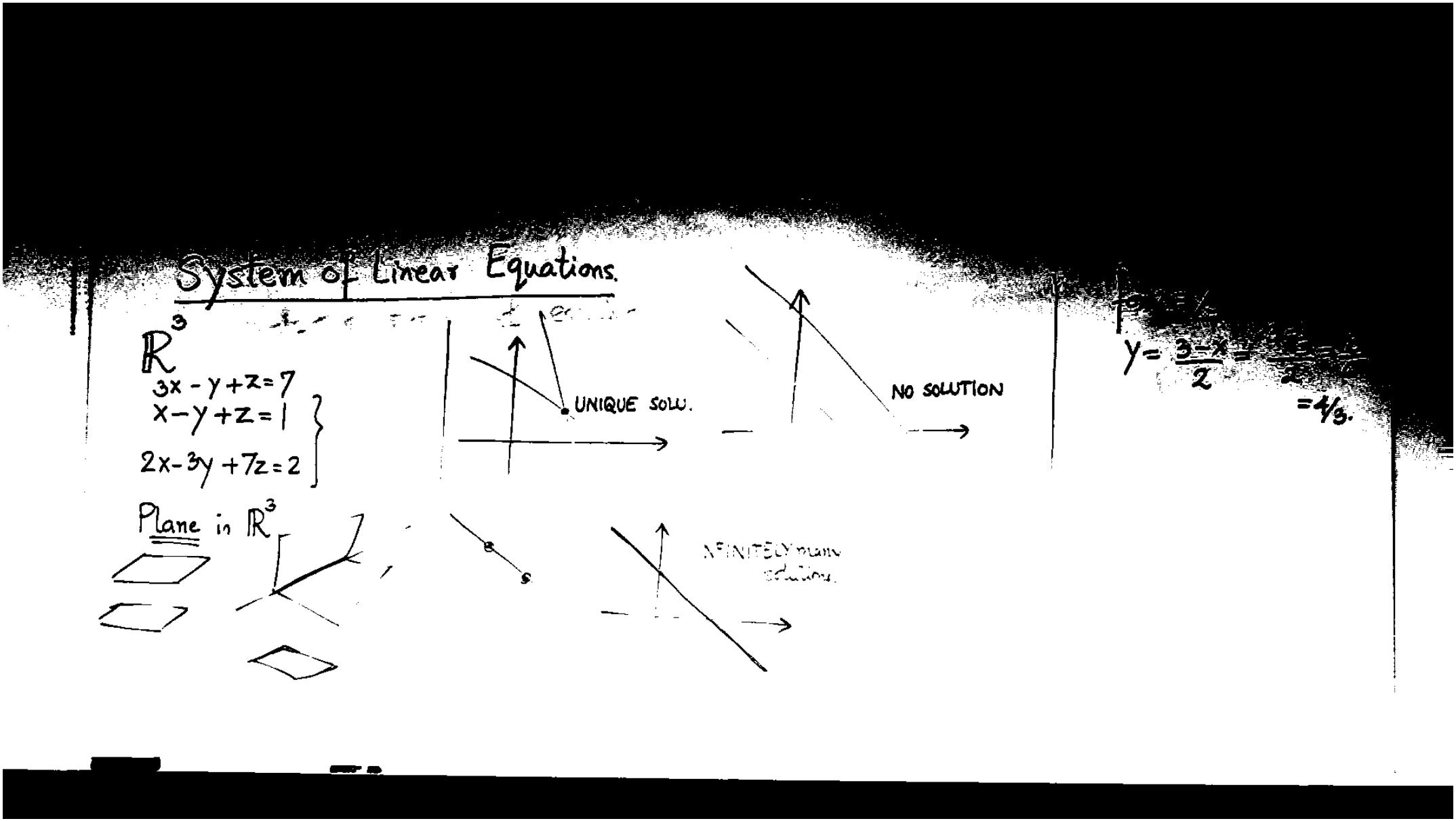
$= 4/3.$

Same full-frame placement on every page. Three frames only; mathematical OCR accuracy was not evaluated.

Otsu global thresholding

Grayscale; Gaussian blur disabled

Real AccessMath2015 | nm_lecture_01_19899 | Native 1920 x 1080 | No smoothing

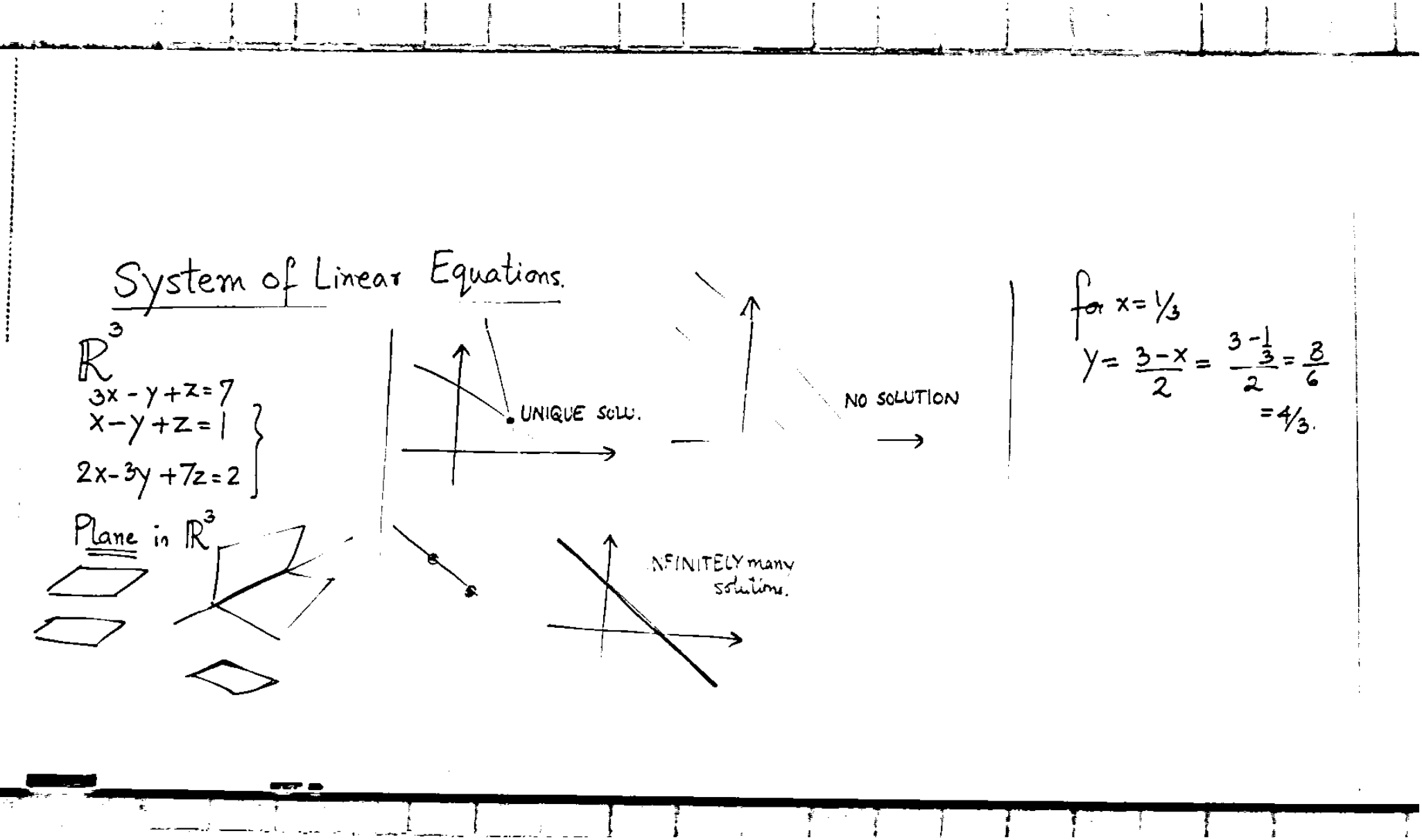


Same full-frame placement on every page. Three frames only; mathematical OCR accuracy was not evaluated.

Sauvola local thresholding

Window = 25; k = 0.2; CLAHE disabled

Real AccessMath2015 | nm_lecture_01_19899 | Native 1920 x 1080 | No smoothing



Same full-frame placement on every page. Three frames only; mathematical OCR accuracy was not evaluated.

LectureNet-guided Sauvola

Window = 25; k = 0.2; dilation radius = 2

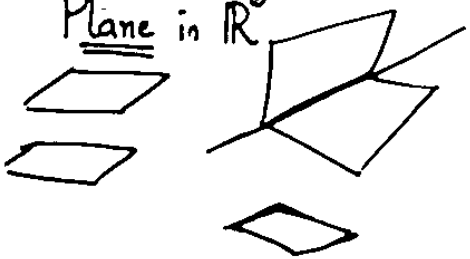
Real AccessMath2015 | nm_lecture_01_19899 | Native 1920 x 1080 | No smoothing

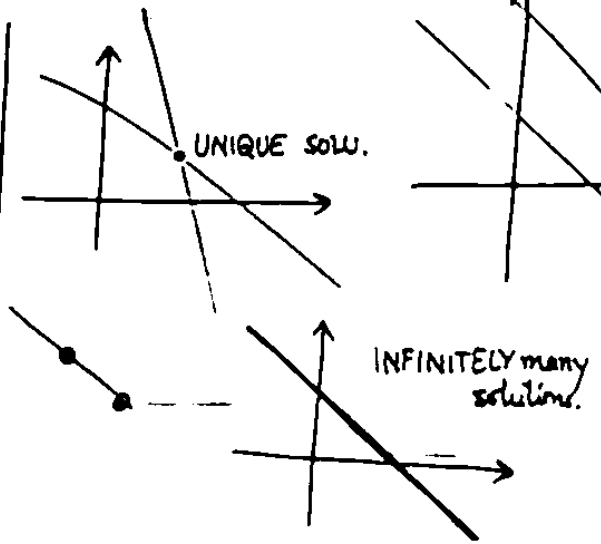
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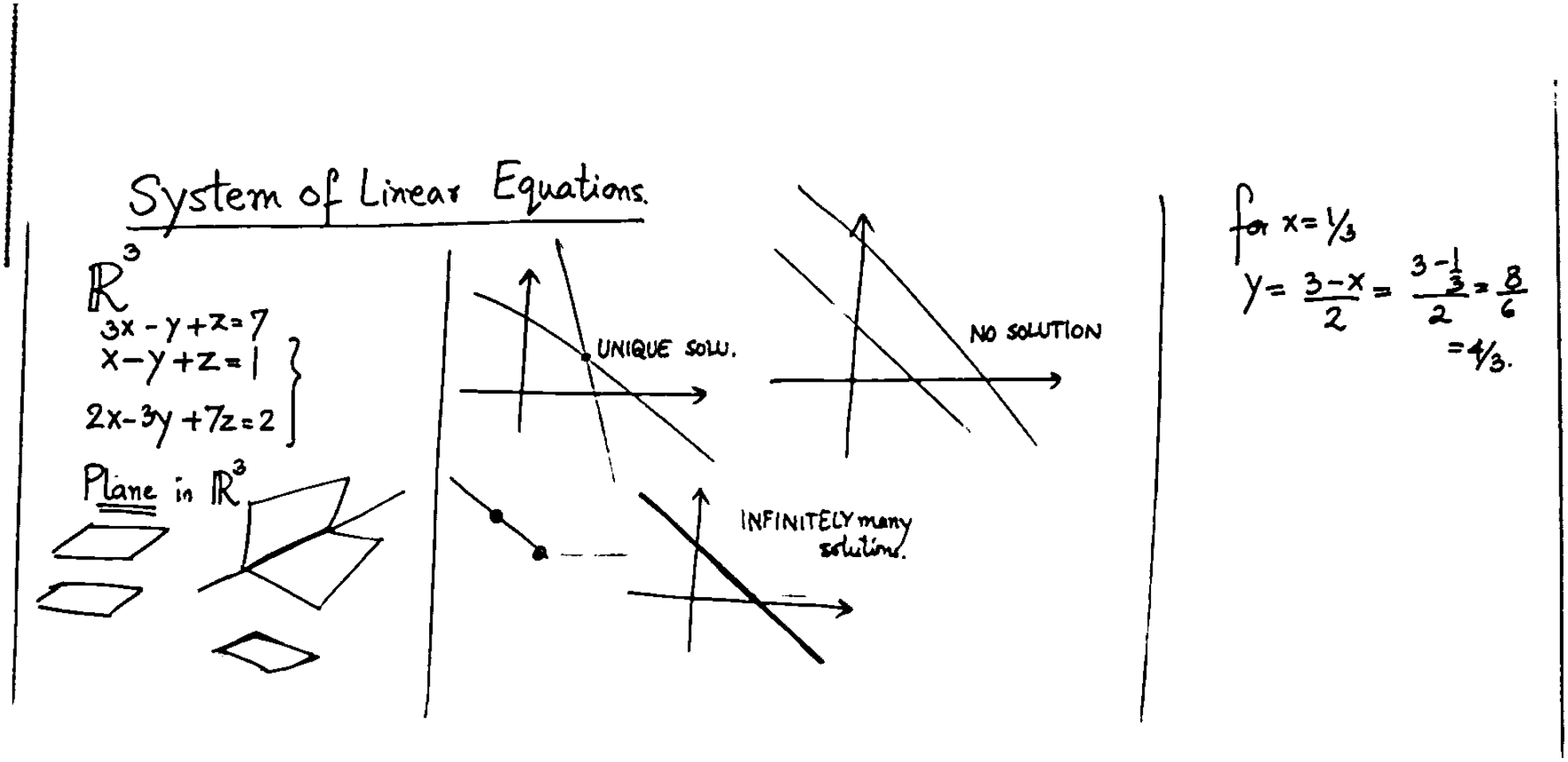
$= 4/3.$

Same full-frame placement on every page. Three frames only; mathematical OCR accuracy was not evaluated.

LectureNet-guided Sauvola + CLAHE

Window = 25; k = 0.2; dilation radius = 2

Real AccessMath2015 | nm_lecture_01_19899 | Native 1920 x 1080 | No smoothing

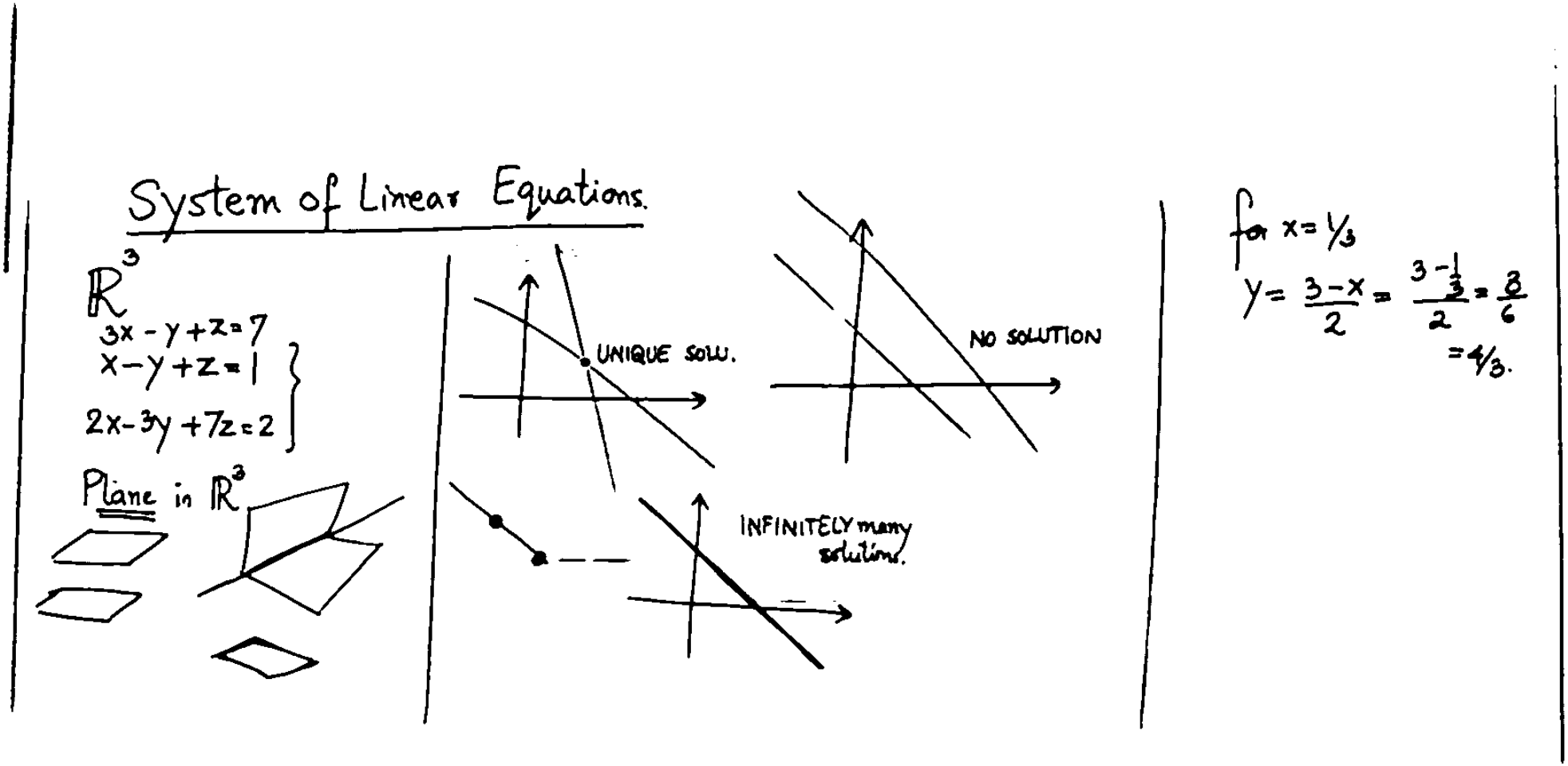


Same full-frame placement on every page. Three frames only; mathematical OCR accuracy was not evaluated.

FCN-LectureNet V34: threshold sweep

Writing probability threshold = 0.35

Real AccessMath2015 | nm_lecture_01_19899 | Native 1920 x 1080 | No smoothing

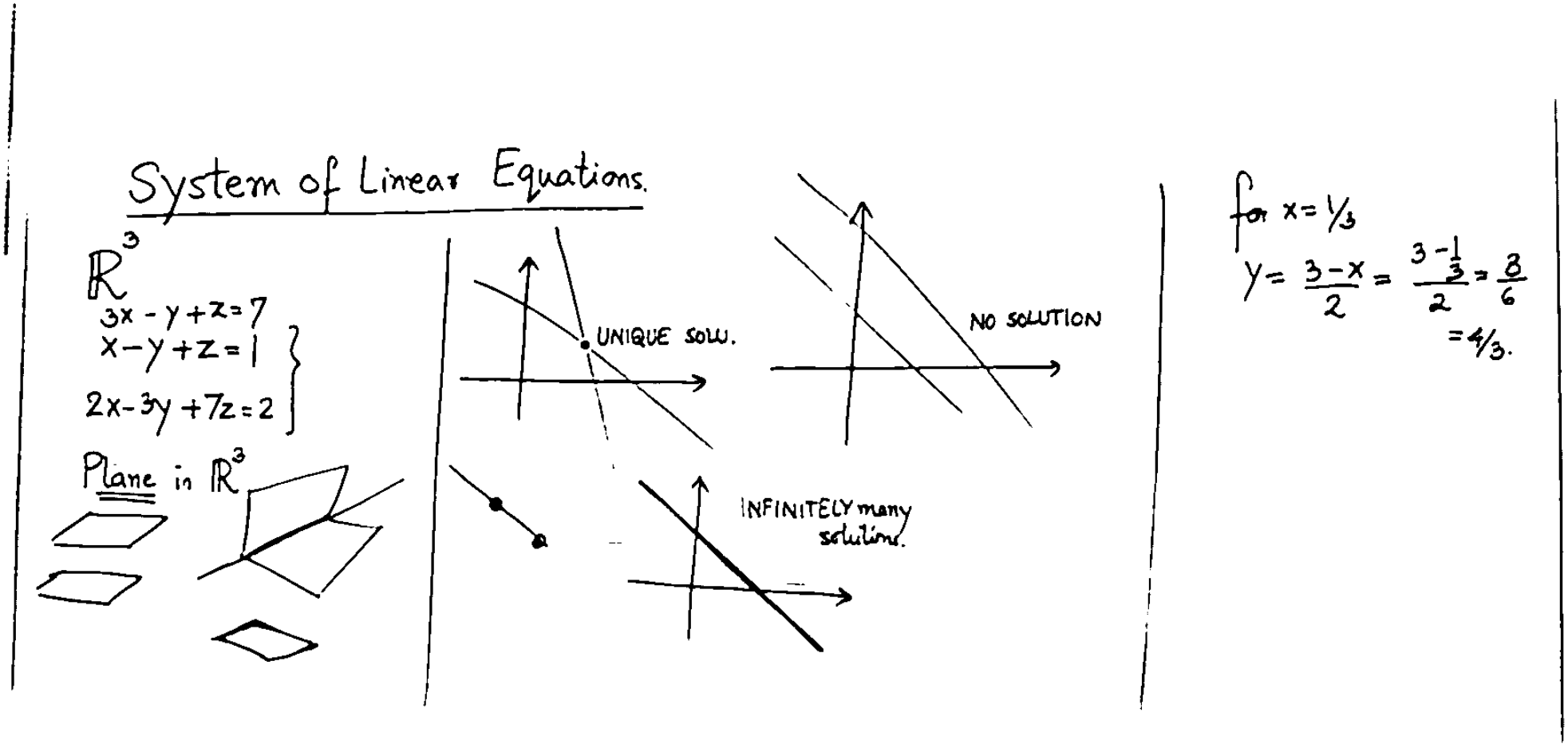


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FCN-LectureNet V34: threshold sweep

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